

Genetic associations between circulating immune cells and periodontitis highlight the prospect of systemic immunoregulation in periodontal care

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Abstract

Periodontitis drives irreversible destruction of periodontal tissue and is prone to exacerbating inflammatory disorders. Systemic immunomodulatory management continues to be an attractive approach in periodontal care, particularly within the context of "predictive, preventive, and personalized" periodontics. In the present study, genetic proxies identified through genome-wide association studies (GWAS) for circulating immune cells and periodontitis were incorporated into a comprehensive Mendelian randomization (MR) framework. Univariable MR, multivariable MR, subgroup analysis, reverse MR, and Bayesian model averaging (MR-BMA) were utilized to investigate the causal relationships. Furthermore, transcriptome-wide association study (TWAS) and colocalization analysis were deployed to pinpoint the underlying genes. Consequently, the MR study indicated a causal association between circulating neutrophils, natural killer T cells, plasmacytoid dendritic cells, and an elevated risk of periodontitis. MR-BMA analysis revealed that neutrophils were the primary contributors to periodontitis. The high-confidence genes *S100A9* and *S100A12*, located on 1q21.3, could potentially serve as immunomodulatory targets for neutrophil-mediated periodontitis. These findings hold promise for early diagnosis, risk assessment, targeted prevention, and personalized treatment of periodontitis. Considering the marginal association observed in our study, further research is required to thoroughly comprehend the biological underpinnings and ascertain the clinical relevance.

eLife assessment

In this **fundamental** study, the authors analyzed associations between circulating immune cells and periodontitis. **Convincing** evidence identifies three immune cell types related to periodontitis, which substantially advances our understanding of periodontitis.

Introduction

Periodontitis imposes a considerable social burden on dental practice and general health

Periodontitis is a highly prevalent disease that affects a considerable percentage of the population. According to large-scale epidemiological research, up to half of all adults worldwide suffer periodontal disease, with severe periodontitis threatening 10.5-12.0 % of them ([Kassebaum et al., 2014](#)). Furthermore, periodontitis is the leading cause of adult tooth loss, necessitating extensive dental procedures such as extractions, dental implants, or prosthetics, which can be costly and time-consuming for both patients and dental practitioners ([Genco and Sanz, 2020](#)). Recent research demonstrated a relationship between periodontitis and inflammatory comorbidities such as type 2 diabetes, cardiovascular disease, rheumatoid arthritis, and inflammatory bowel disease ([Hajishengallis and Chavakis, 2021](#)). The high prevalence and harmful implications of periodontitis underline the importance of managing periodontitis to maintain oral and general health ([Peres et al., 2019](#)). Since early-stage prevention is the most significant way to improve health, the identification of additional potential risk factors was required to provide predictive, preventive, and personalized strategies for periodontal care ([Ma et al., 2021](#)).

Evidence from epidemiology and pathophysiology demonstrates the impact of circulating immune cells on periodontitis

Periodontitis is a chronic inflammatory disease characterized by the interactions between microorganisms and host immune response ([Curtis et al., 2020](#)). The immune response to periodontitis comprises both innate and adaptive immunity, with multiple cytokines, immune cells, and inflammatory pathways participating in a complex interplay ([Dutzan et al., 2016](#)). Systemic immunological alternations, such as circulating immune cells, play a crucial role in the initiation and progression of periodontitis ([Cekici et al., 2014](#)). An observational study indicated that patients with periodontitis experience a greater level of circulating leukocytes ([Noz et al., 2021](#)), while another discovered that the distribution of B cells alters in the context of severe periodontitis, with a higher proportion of circulating memory B cells ([Demoersman et al., 2018](#)). Furthermore, inflamed periodontal tissue recruits immune cells from circulation ([Hajishengallis, 2020](#)). With the progression of periodontitis, there is a significant alteration in the quantity of immune cells present within the periodontal tissue. Specifically, an increase in the count of both monocytes and B cells is observed, whereas a decrease is noted in the count of T cells. ([Nair et al., 2014](#); [Steinmetz et al., 2016](#)). The promising concept of "trained immunity" has recently provided a greater understanding of the host immune response in periodontitis ([Netea et al., 2020](#)), which can explain the fact that the increased hyper-responsiveness of circulating immune cells from patients with periodontitis as well as its probable mechanism of mediating periodontitis and its comorbidities ([Li et al., 2023](#)).

Immunomodulation of systemic immune response serves as a hub for periodontal care

Systemic immunomodulation management has the potential to improve host homeostasis by altering the composition and function of the immune milieu ([Yang et al., 2021](#)). Periodontitis can be effectively managed by restricting immune cell activation, implying that immunomodulators have significant promise in constructing comprehensive strategies for periodontal management ([Zidar et al., 2021](#)). For example, resveratrol, quercetin, and N-acetylcysteine were reported to reduce the release of reactive oxygen species (ROS) by neutrophils, which aided in the prevention of periodontitis ([Orihuela-Campos et al., 2015](#)). Nonetheless, from a medical and therapeutic perspective, it is critical to determine whether the link between circulating immune cells and periodontitis is merely correlative or driven by causative mechanistic interactions ([Lamont and Hajishengallis, 2015](#)). Understanding the role of systemic immune alternations in periodontitis is critical for developing an effective strategy for early screening of high-risk patients, prompt implementation of definitive prevention, and individualized deployment of targeted treatment, all with the goal of reducing unexpected inflammatory responses, maintaining oral health, and avoiding complications ([Zhang et al., 2023](#)).

Mendelian randomization provides a potent complement to causal inference in terms of genetics

Previous research has substantiated the potential of immunomodulation management in predicting and preventing periodontitis; however, in observational studies, the association is frequently disguised by reverse causality, confounding factors, and disease conditions, which obscured the intrinsic causal inference between them ([Hajishengallis and Korostoff, 2017](#)). Mendelian randomization (MR) investigates the causal relationships between risk factors and diseases by exploiting genetic variants as instrumental variables (IVs) ([Davies et al., 2018](#)), which is less likely to be affected by underlying bias or disease condition, in that alleles are randomly allocated from parents to offspring ([Julian et al., 2023](#)). Notably, MR with distinct causal relationships may provide fresh evidence from a genomics perspective ([Golubnitschaja et al., 2014](#)). We postulate that individuals with a disproportionate immunological network have a higher risk of periodontitis due to unexpected inflammatory reactions.

Methods

Study design

The present study, as shown in [Figure 1](#), was based on the Strengthening the Reporting of Observational Studies in Epidemiology using the Mendelian Randomization (STROBE-MR) checklist ([Skrivankova et al., 2021](#)). The present research aims to evaluate the causal association between circulating immune cells and the risk of periodontitis, providing insight into opportunity for systemic immunomodulation management in periodontal care. We utilized summary statistics from publicly available genome-wide association studies (GWAS) to perform both univariable MR (UVMR) and multivariable MR (MVMR) analyses. Furthermore, we replicated the UVMR analysis by excluding potentially pleiotropic SNPs, and subsequently conducted subgroup and reverse MR analyses. Additionally, the Bayesian model averaging (MR-BMA) was employed to pinpoint the predominant characteristics with causal signals. Finally, we conducted a transcriptome-wide association study (TWAS) and colocalization analysis to discern potential genes implicated in biological relationships.

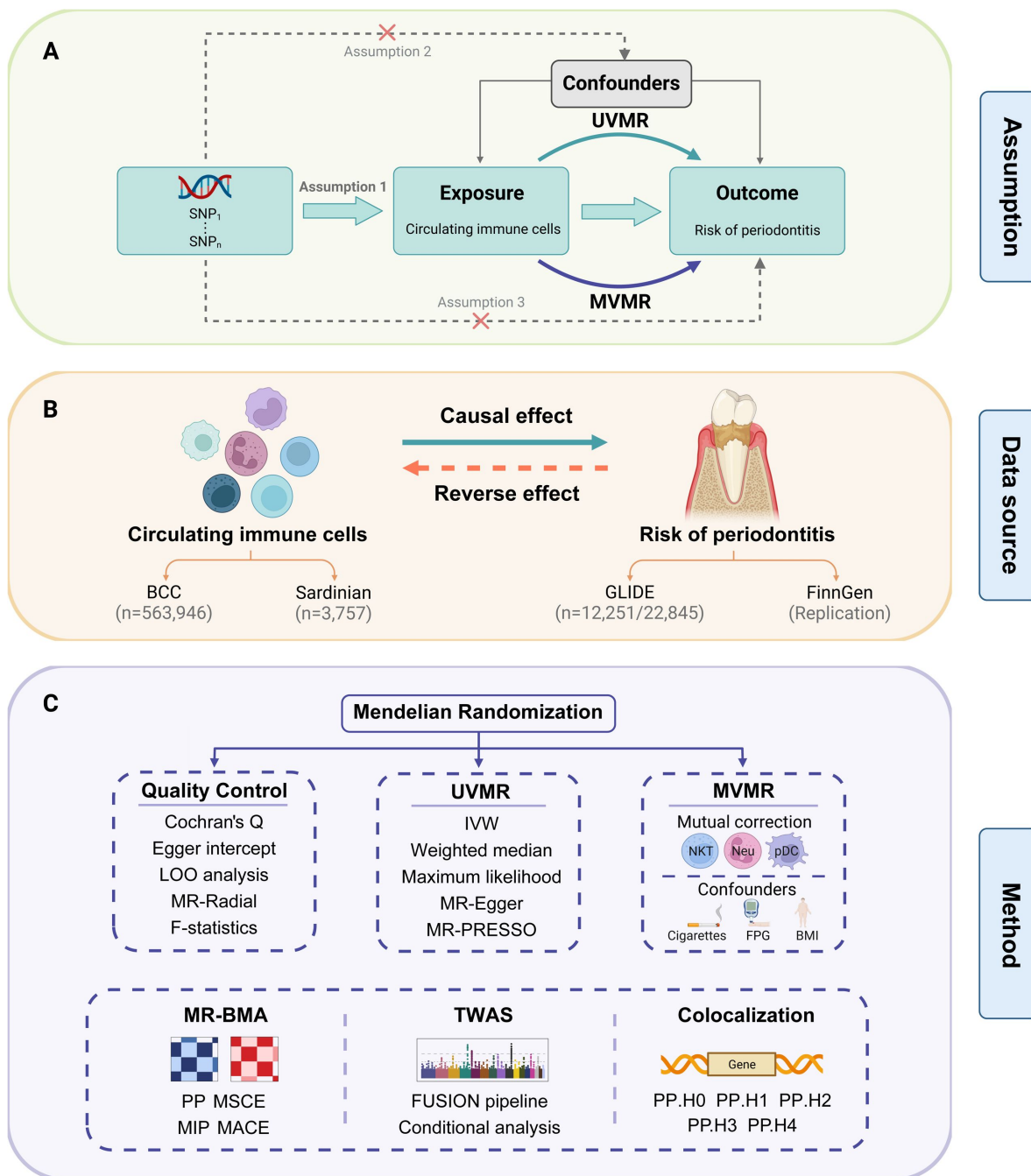


Figure 1.

Study design. **(A)** Overview of the process and principal assumptions of MR. **(B)** Data sources of the GWASs. **(C)** Methods performed in the present study.

Abbreviations and Notes: BCC, Blood Cell Consortium; BMA, Bayesian model averaging, a high-throughput method based on non-linear regression; BMI, body mass index; FPG, fasting plasma glucose; FUSION, functional summary-based imputation; GLIDE, Gene-Lifestyle Interactions in Dental Endpoints collaboration consortium; GWAS, genome-wide association study; IVW, inverse variance weighted, the primary method in MR to explore the association between exposure and outcome; LOO, leave-one-out, a method for detecting potential influential SNPs; SNP, single nucleotide polymorphism, as genetic instrumental variables for the exposure and outcome; MACE, model-averaged causal estimate; MIP, marginal probability of inclusion; MR, Mendelian randomization; MR-PRESSO, Mendelian Randomization Pleiotropy RESidual Sum and Outlier, a method for assessing and rectifying pleiotropic SNPs; MSCE, model-specific causal estimate; MVMR, multivariable Mendelian randomization, a MR model for adjusting confounding and mutual correction; Neu, neutrophil; NKT, Natural Killer T cell; pDC, plasmacytoid dendritic cell; PP, posterior probability; TWAS, transcriptome-wide association study; UVMR, univariable Mendelian randomization.

Data source

Summary-level data on 17 circulating immune cells were obtained from the large-scale GWAS conducted by the Blood Cell Consortium (BCC) and the Sardinian cohort ([Orrù et al., 2020](#); [Vuckovic et al., 2020](#)). The GWAS data for periodontitis and its subtypes were supplied by the Gene-Lifestyle Interactions in Dental Endpoints collaboration (GLIDE) consortium and the FinnGen cohort ([Kurki et al., 2023](#); [Shungin et al., 2019](#)). To maintain the homogeneity within the target group and minimize overlap, we performed a screening process on the population. Specifically, the population from the Latin American and the UK Biobank cohort was excluded from the periodontitis data set ([Ye et al., 2023](#)). The features of GWAS characteristics and included cohorts were highlighted in [Table 1](#) and [Supplementary File 1 — Table s1](#).

Candidates for IVs underwent a thorough set of screening procedures. To equalize the sample disparities among databases, a complicated screening criterion was performed. We initially filtered the p -values of the single nucleotide polymorphisms (SNPs), followed by the selection of independent SNPs using the linkage disequilibrium (LD) approach. The R^2 and F-statistics were introduced to demonstrate the degree of genetic variation explained and their relative impact on the outcomes, and SNPs with F-statistics < 10 would be removed based on the first MR assumption ([Papadimitriou et al., 2020](#)). In addition, SNPs that exhibited a direct association with the outcome would also be deleted to support the third MR assumption. Throughout the harmonization processes, palindromic and ambiguous SNPs were eliminated to ensure the reliability and validity of causal inference. In MVMR, we excluded SNPs situated in the major histocompatibility complex area (MHC, 6p21.31) due to its complexity and confounding effects ([Burgess and Thompson, 2015](#)).

Univariable and multivariable Mendelian randomization

In UVMR, the inverse-variance weighted (IVW) method was performed as the primary analysis, and four alternative MR methods, including weighted median, maximum likelihood, MR-Egger, and MR pleiotropy residual sum and outlier (MR-PRESSO) global test were employed for sensitivity testing to assess the robustness of the IVW estimates. Briefly, the IVW assumes that all genetic variations meet the conditions and integrated estimates from multiple genetic variants by weighting them inversely to variances ([Sanderson et al., 2022](#)). The weighted median generates precise estimates when more than half of the SNPs are valid ([Gormley et al., 2023](#)). The maximum likelihood offers a normal bivariate distribution to estimate causal effects by maximizing the likelihood function with a linear relationship ([Xue et al., 2021](#)). MR-Egger provides estimates after accounting for possible horizontal pleiotropy discovered by its incorporated intercept test, albeit the estimates were frequently underpowered ([Bowden et al., 2016](#)). MR-PRESSO detects outliers that cause pleiotropy and generates estimates once these outlier SNPs are eliminated ([Verbanck et al., 2018](#)). The observed significant results were considered "robust" if the effect of sensitivity analyses was identical to that of IVW method, yielding a p -value < 0.05.

To gauge the individual influence of each variant, MVMR analysis with mutual adjustment was performed, followed by a correction for associated confounders ([Burgess and Thompson, 2015](#)). We have incorporated covariates, including the number of cigarettes smoked, fasting plasma glucose (FPG) levels, and body mass index (BMI) into the MVMR analysis, given that these factors could indirectly affect systemic immune responses and inflammation ([Liu et al., 2023](#)). The MVMR-IVW method was utilized as the primary test, supplemented by the MVMR-least absolute shrinkage and selection operator (MVMR-LASSO) and the MVMR-Egger method ([Bowden et al., 2016](#)). The MVMR-LASSO regression yields dependable estimations for moderate-to-high degrees of heterogeneity or pleiotropy, and it also assists in alleviating the potential effects of multicollinearity among variables ([Grant and Burgess, 2021](#)).

Phenotype	Year	Sample size (ncase/ncontrol)	nSNP	Ancestry	Unit	Consortium/cohort	PMID
Exposure							
Circulating immune cells	2020	563,946	15million	European	/nL	BCC	32888494
Lymphocyte subsets	2020	3,757	15.2million	European	/ug	Sardinian cohort	32929287
Outcome							
Periodontitis	2019	35,096 (12,251/22,845)	10.8million	European	Event	GLIDE	31235808
Chronic periodontitis (FinnGen)	2023	263,668 (4,434/259,234)	20.2million	European	Event	FinnGen (R9K11)	36653562
Chronic gingivitis (FinnGen)	2021	196,245 (850/195,395)	16.4million	European	Event	FinnGen (R5K11)	
Gingival hyperplasia (FinnGen)	2023	259,613 (379/259,234)	20.2million	European	Event	FinnGen (R9K11)	
Covariate							
Cigarettes smoked per day	2019	249,752	12million	European	1/SD	GSCAN	30643251
Fasting plasma glucose	2021	200,622	31million	European	mmol/L	MAGIC	34059833
Body mass index	2018	681,275	2.3million	European	kg/m ²	GIANT	30124842

Abbreviations: BCC, Blood Cell Consortium; GIANT, GWAS of the Genetic Investigation of ANthropometric Traits; GLIDE, Gene-Lifestyle Interactions in Dental Endpoints; GSCAN, GWAS and Sequencing Consortium of Alcohol and Nicotine use; MAGIC, Meta-Analysis of Glucose and Insulin-related Traits Consortium; SD, standard deviation; SNP, single nucleotide polymorphism.

Table 1.

Characteristics of the GWAS data for MR.

The heterogeneity and horizontal pleiotropy of the results were quantified using Cochran's Q statistics and the intercept term in MR-Egger regression, respectively. The MR-Radial, a more sensitive method for outliers, would be employed to detect and remove outlier SNPs whenever heterogeneity or pleiotropy was discovered ([Bowden et al., 2018](#)). The leave-one-out (LOO) analysis and scatter plot were carried out to detect influential SNPs.

Bayesian model averaging

As a multivariate framework for high-throughput risk factors based on non-linear regression, the MR-BMA was then employed to explore the leading traits responsible for outcome ([Zuber et al., 2020](#)). First, we used closed-form Bayes factors and independence priors to calculate the posterior probability (PP) and model-specific causal estimates (MSCE) of each variant. Next, the total PPs for all potential models were added up to determine the marginal probability of inclusion (MIP). The model-averaged causal estimate (MACE), which reflected the average direct effect of each metabolic trait on the outcomes, was also used to compare risk factors and interpret the directions. Finally, the best model was chosen preferably based on the ranking of each model's MIP and PP values. The Q-statistic and Cook's distance were used to identify invalid outliers and influential variants within the model. A genetic variant was defined as either an outlier or an influential variant if it possessed a q-value exceeding 10 or if its Cook's distance surpassed the median of the corresponding F-distribution. The MR-BMA would be repeated once unqualified variations were discovered ([Eledum, 2021](#)).

Transcriptome-wide association study

We exploited the updated Genotype-Tissue Expression (GTEx) project Version 8 whole-blood data for transcriptome-wide association study (TWAS) analysis ([Gusev et al., 2016](#)). First, the functional summary-based imputation (FUSION) pipeline was used to infer the transcriptome associated with significant outcomes, among which the optimal gene expression model was chosen by comparing the values of R^2 provided by Bayesian sparse linear mixed models and multiple penalized linear regressions. A Bonferroni-corrected criterion of $P < 6.27 \times 10^{-6}$ (0.05/7,890 genes) was adopted as a measure of statistical significance. Then, conditional analysis and permutation testing were implemented to assess the dependability and robustness of the gene transcript-trait relationships discovered through TWAS. Finally, we performed a expression quantitative trait locus (eQTL) colocalization analysis on TWAS-derived genes to determine whether the association was caused by a single causal SNP (PP.H4) or distinct causal SNPs (PP.H3). $PP.H3 + PP.H4 > 0.8$ was considered significant evidence of colocalization ([Wallace, 2020](#)).

Statistical analyses

Odds ratios (OR) with 95% confidence intervals (CI) were utilized to measure the influence of circulating immune cells on the likelihood of periodontitis. All statistical analyses were performed using "TwoSampleMR" (version 0.5.7), "MRPRESSO" (version 1.0), "RadialMR" (version 1.1), "coloc" (version 5.1.0.1), and "MVMR" (version 0.4) packages in R software (version 4.3.1).

Results

Selection of instrumental variables

A rigorous threshold of $P < 1 \times 10^{-9}$ was applied to the database with an abundance of positive SNPs (as in the BCC consortium) to ensure the reliability of IVs. Otherwise, a relatively strict standard of $P < 1 \times 10^{-6}$ was initially adopted (as in the Sardinian cohort), and we would loosen it at $P < 5 \times 10^{-6}$ if less than three SNPs met this threshold (a basic requirement for MR-PRESSO analysis). As a result, a total of 1940 SNPs were selected as IVs in the present study ([Supplementary File 1](#) — Table s2). The F-statistics ranged from 28.67 to 220.07, indicating a low risk of weak instrument bias.

Univariable Mendelian randomization

Three circulating immune cells were identified to be suggestively significant in the IVW method [OR: 1.09, 95% CI: 1.01-1.17, $P = 0.030$ for Natural Killer T cells (NKT); OR: 1.11, 95% CI: 1.00-1.23, $P = 0.042$ for neutrophils; OR: 1.13, 95% CI: 1.02-1.25, $P = 0.025$ for plasmacytoid Dendritic Cells (pDC)], which were further supported by the maximum likelihood and MR-PRESSO ([Figure 2A,2B](#), [Supplementary File 1](#) — Table s3). The MR-Egger regression revealed no evidence of horizontal pleiotropy (p -values for intercept > 0.05). However, significant heterogeneity was detected in two traits (memory B cell and monocyte) ([Supplementary File 1](#) — Table s4), which faded after the removal of outliers ([Supplementary File 1](#) — Table s5, [Supplementary File 2](#) — Figure s1). Moreover, the LOO analysis showed no influential SNPs that were significantly linked with the outcome ([Supplementary File 2](#) — Figure s2). The observed significant results remained robust after removing pleiotropic SNPs ([Supplementary File 1](#) — Table s6, [Supplementary File 2](#) — Figure s3), and the scatter plot displayed a balance distribution amongst SNPs ([Figure 2C-2E](#)).

Multivariable Mendelian randomization

After accounting for variable mutual adjustment and covariate correction for potential confounders, the causal relationship between circulating neutrophils and periodontitis remained stable with no evidence of heterogeneity or pleiotropy. NKT remained stable after modifying pDC and BMI, owing to the considerable heterogeneity of its impact on BMI; nonetheless, while the MR-LASSO analysis was used to make appropriate corrections, it was still advised to exercise caution when interacting with the effect. What's more, pDC maintained stability after adjusting for NKT, whereas the strength of the observed association was compromised in the MR-Egger sensitivity analysis ([Figure 3](#), [Supplementary File 1](#) — Table s7).

Bayesian model averaging

The best risk models and factors were ordered and prioritized based on their PP and MIP ([Table 2](#), [Supplementary File 1](#) — Table s8). Consequently, we observed that neutrophil was the best model and leading factor for periodontitis (PP=0.771, MSCE=0.108, MIP=0.895, MACE=0.097), followed by NKT and pDC. The Cochran's Q test and Cook's distance failed to detect outlier or influential variations ([Supplementary File 2](#) — Figure s4).

Subgroup analysis and reverse Mendelian randomization

A replication UVMR of three subgroups of periodontal diseases (chronic periodontitis, chronic gingivitis and gingival hyperplasia) was also performed in the FinnGen cohort ([Figure 4](#), [Supplementary File 1](#) — Table s9-11). The significant findings identified within the GLIDE database, however, were not replicated during subgroup analysis, which may be attributed to the heterogeneity of periodontal disease, as well as variations in the population composition across datasets. Intriguingly, B cell was discovered to be involved in the subgroups of the FinnGen population (OR: 1.11, 95% CI: 1.02-1.22, $P = 0.019$ for chronic periodontitis; OR: 1.39, 95% CI: 1.02-1.88, $P = 0.036$ for gingival hyperplasia). Reverse MR revealed no indication of reverse causality ([Figure 4](#), [Supplementary File 1](#) — Table s12).

Transcriptome-wide association study

The TWAS indicated that five cross-trait genes, including *CC2D2B* (10q24.1), *RP11-326C3.7* (11p15.5), *USP3* (15q22.31), *HERC1* (15q22.31), and *AMFR* (16q13), may be implicated in the interaction of circulating immune cells with periodontitis ([Figure 5A, B](#)). After Bonferroni correction ($P < 6.27 \times 10^{-6}$), we identified 658 of 3081 characteristics that were significantly associated with neutrophils, 5 of 443 with NKT, and 5 of 1038 with pDC. Within a broad criteria ($P < 5 \times 10^{-4}$), we discovered 6 of 423 characteristics were linked to periodontitis ([Figure 5C](#)).

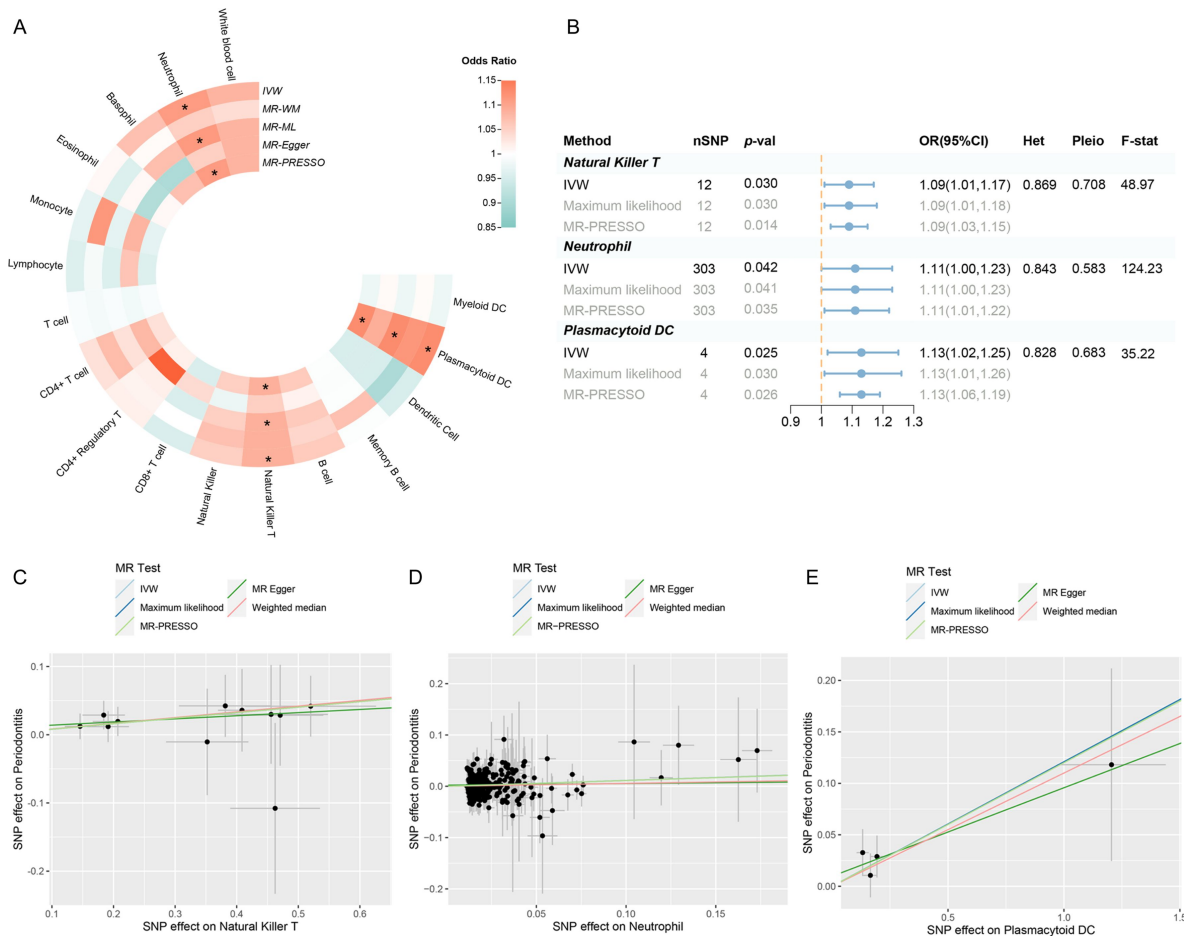


Figure 2.

Results of the UVMR. **(A)** A circular heatmap representing the MR analyses for the associations between circulating immune cells and the risk of periodontitis. Lines, from outermost to innermost, represent IVW, MR-WM, MR-ML, MR-Egger, and MR-PRESSO respectively. The color scale of the heatmap is based on the OR. * $P < 0.05$. **(B)** A forest plot of the MR analyses for significant results in **Figure 2A** ($P < 0.05$). The effects are quantified using OR with 95% CI. **(C-E)** The effect estimate for each individual variant in Natural Killer T cell **(C)**, neutrophil **(D)**, and plasmacytoid DC **(E)** is provided by plotting SNP-outcome associations against SNP-exposure associations. The regression slope fitted by different MR methods is represented by lines with different colors.

Abbreviations: CI, confidence interval; DC, dendritic cell; F-stat, F-statistic; IVW, inverse variance weighted; Het, heterogeneity; MR, Mendelian randomization; MR-ML, Mendelian randomization weighted median; MR-WM, Mendelian randomization maximum likelihood; MR-PRESSO, Mendelian Randomization Pleiotropy RESidual Sum and Outlier; OR, odds ratio; Pleio, pleiotropy; SNP, single nucleotide polymorphism; UVMR, univariable Mendelian randomization.

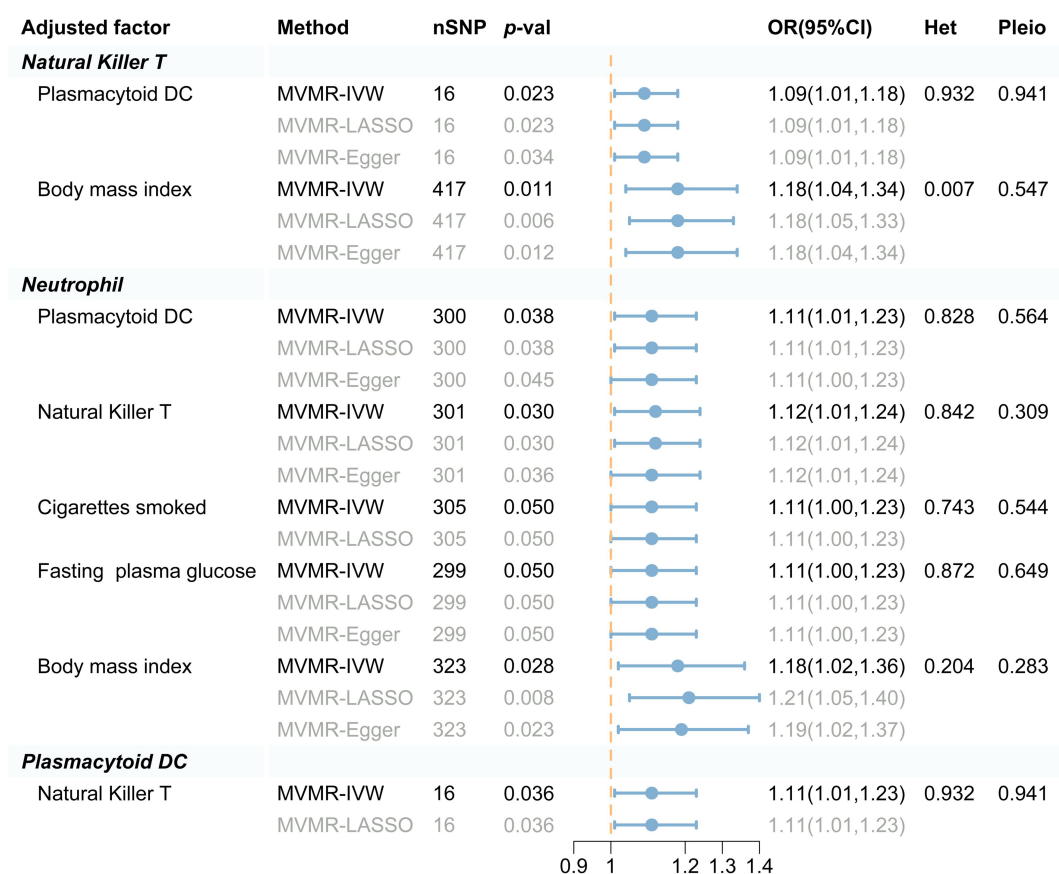


Figure 3.

Forest plot analysis of the MVMR of significant results following mutual adjustment and confounder correction. The effects are quantified using OR with 95% CI.

Abbreviations: CI, confidence interval; DC, Dendritic Cell; IVW, inverse variance weighted; Het, heterogeneity; LASSO, least absolute shrinkage and selection operator; MVMR, multivariable Mendelian randomization; OR, odds ratio; Pleio, pleiotropy; SNP, single nucleotide polymorphism.

Trait	Ranking by MIP	MIP	MACE	Ranking by PP	PP	MSCE
Neutrophil	1	0.895	0.097	1	0.771	0.108
Natural Killer T cell	2	0.135	-0.003	2	0.056	-0.017
Plasmacytoid DC	3	0.102	0	3	0.045	0.007

Abbreviations: DC, dendritic cell; MACE, average causal effect of risk factor model; MIP, marginal inclusion probability; MR-BMA, Mendelian randomization-Bayesian model averaging; MSCE, model-specific causal estimates; PP, poster probability.

Table 2.

Ranking of risk factors and models for periodontitis in MR-BMA analysis.

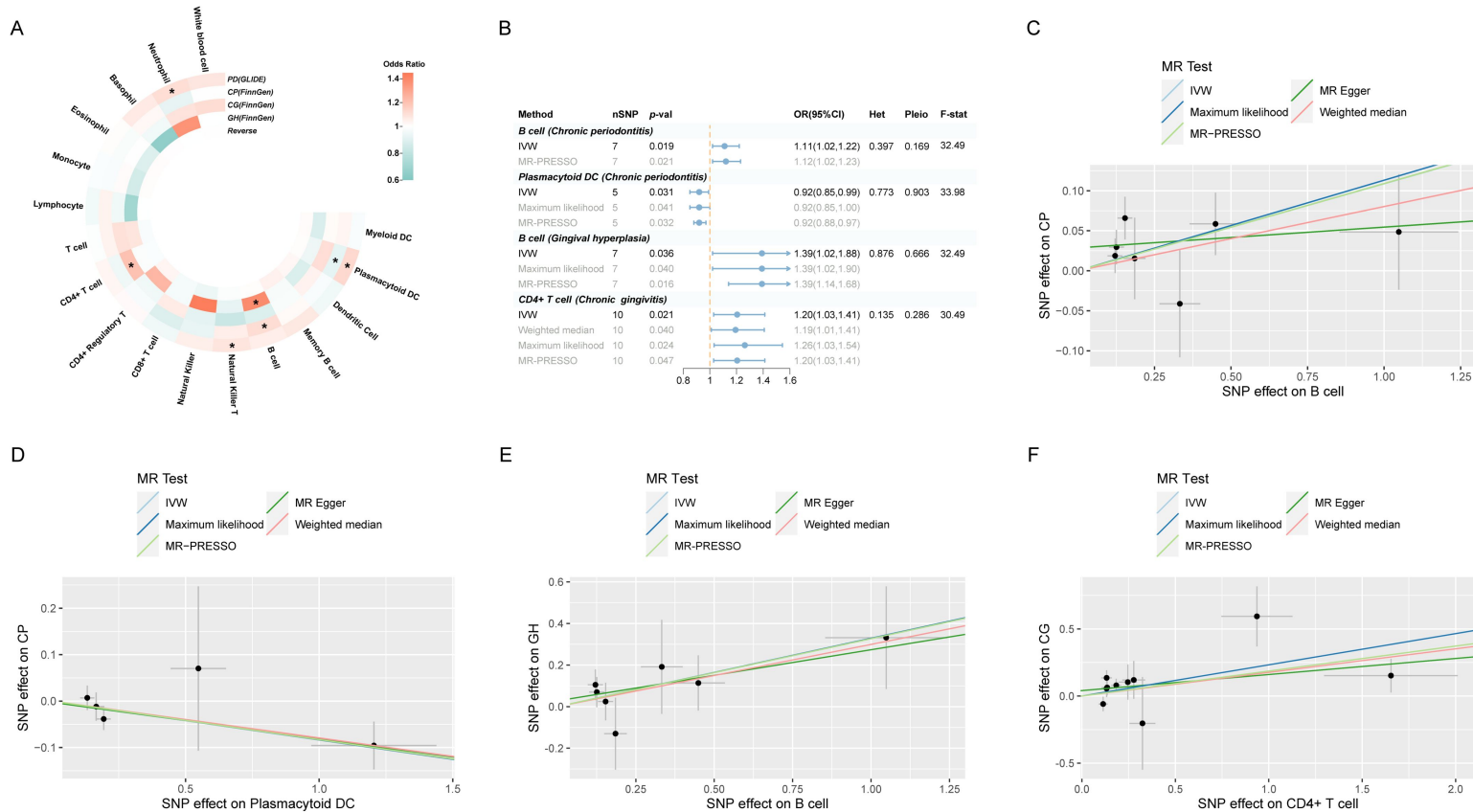


Figure 4.

Results of the subgroup and reverse MR. **(A)** A circular heatmap illustrates the results of the subgroup analysis and reverse MR. Lines in the heatmap represent periodontitis (GLIDE), chronic periodontitis (FinnGen), chronic gingivitis (FinnGen), gingival hyperplasia (FinnGen), and reverse MR analysis, progressing from outside to inside. The color scale of the heatmap is determined by the odds ratio (OR). * $P < 0.05$. **(B)** A forest plot of the MR analyses for significant results in **Figure 4A** ($P < 0.05$). The effects are quantified using OR with 95% CI. **(C-F)** The effect estimates for each individual variant in B cell (CP), plasmacytoid DC (CP), B cell (GH) and CD4+ T cell (CG) is provided by plotting SNP-outcome associations against SNP-exposure associations. The regression slope fitted by different MR methods is represented by lines with different colors. **Abbreviations:** CG, chronic gingivitis; CI, confidence interval; CP, chronic periodontitis; DC, dendritic cell; F-stat, F-statistic; GH, gingival hyperplasia; GLIDE, Gene-Lifestyle Interactions in Dental Endpoints collaboration consortium; Het, heterogeneity; IVW, inverse variance weighted; MR-PRESSO, Mendelian Randomization Pleiotropy RESidual Sum and Outlier; OR, odds ratio; PD, periodontitis; Pleio, pleiotropy; SNP, single nucleotide polymorphism.

Supplementary File 2 — Figure s5). Notably, four of these high-confidence genes were discovered to be involved with multiple phenotypes: *S100A9*, *S100A12* (neutrophils and periodontitis); *MCM6*, *P14KAP2* (neutrophils and pDC) (**Table 3**, **Figure 5D**). Most of these significant features survived both conditional analysis and permutation testing (381/658 for neutrophils, 3/5 for NKT, 5/5 for pDC, and 5/6 for periodontitis). The majority of them were shown to be colocalized with their respective phenotype (554/658 for neutrophils, 4/5 for NKT, 3/5 for pDC, 0/6 for periodontitis), implying that shared and pleiotropic SNPs influence both gene expression and phenotype (**Supplementary File 1** — Table s13-16).

Discussion

Summary of key findings

In the present research, we employed MR to explore the potential links between circulating immune cells and periodontitis. Our study revealed causal relationships between elevated levels of circulating neutrophils, Natural Killer T cells, and plasmacytoid Dendritic Cells with a higher risk of periodontitis. TWAS and colocalization analysis demonstrated possible high-confidence and cross-trait genes to be engaged in their interaction.

Circulating neutrophils play a significant part in periodontitis and inflammatory comorbidities

Notably, our findings suggested that circulating neutrophils may play a leading causal role in the likelihood of periodontitis, and it remained robust after correcting for potential confounding factors and outliers. Neutrophils are abundant and short-lived myeloid cells that can be rapidly recruited to inflammatory sites, serving as the first line of defense against infections and other host insults. In recent years, a profound understanding of the role of neutrophils in chronic inflammatory diseases has been gained, where they may directly act as effectors of destructive inflammation. However, a scarcity of neutrophils can also trigger destructive tissue inflammation, their pivotal role in maintaining physiological equilibrium ([Ley et al., 2018](#)). Numerous pieces of clinical evidence have uncovered that neutrophils account for a significant portion of inflammatory tissue damage and that the severity of periodontitis is positively correlated with the overproduction, dysregulation, or hyperactivity of neutrophils ([Chapple et al., 2023](#); [Fine et al., 2021](#)). A case-control study indicated that periodontitis patients suffered from a higher level of apoptotic circulatory neutrophils than healthy people ([Nicu et al., 2018](#)). An increased neutrophil count could suggest the inflammatory burden of gingivitis and dental plaque in the oral cavity ([Sreenivasan and Prasad, 2022](#)). Another study discovered that neutrophil depletion ameliorated experimental periodontitis while unrestrained recruitment aggravated it ([Dutzan et al., 2018](#)).

Neutrophil-mediated inflammatory responses are crucial in the pathogenesis of periodontitis, influencing not only its localized manifestation but also its systemic complications, thereby associating it with a broad spectrum of inflammatory complications ([Hajishengallis and Chavakis, 2021](#)). During periodontitis, neutrophils act as the primary defense against bacteremia, with their primary function being to engulf and eradicate these bacteria, thus inhibiting their dissemination within the body. Following this, neutrophils liberally release pro-inflammatory cytokines in the connective tissue adjacent to the periodontal pocket. It is plausible that these cytokines could permeate into the bloodstream, thereby potentially exacerbating systemic inflammation ([D'Aiuto et al., 2013](#)). In severe periodontitis cases, patients experience low-grade systemic inflammation, as indicated by elevated levels of neutrophils and pro-inflammatory mediators, compared to their healthy counterparts ([Schenkein et al., 2020](#)). Preclinical research also indicates that ligature-induced periodontitis (LIP) is accompanied by an elevation in circulating neutrophil counts, resulting in endothelial dysfunction and vascular inflammation

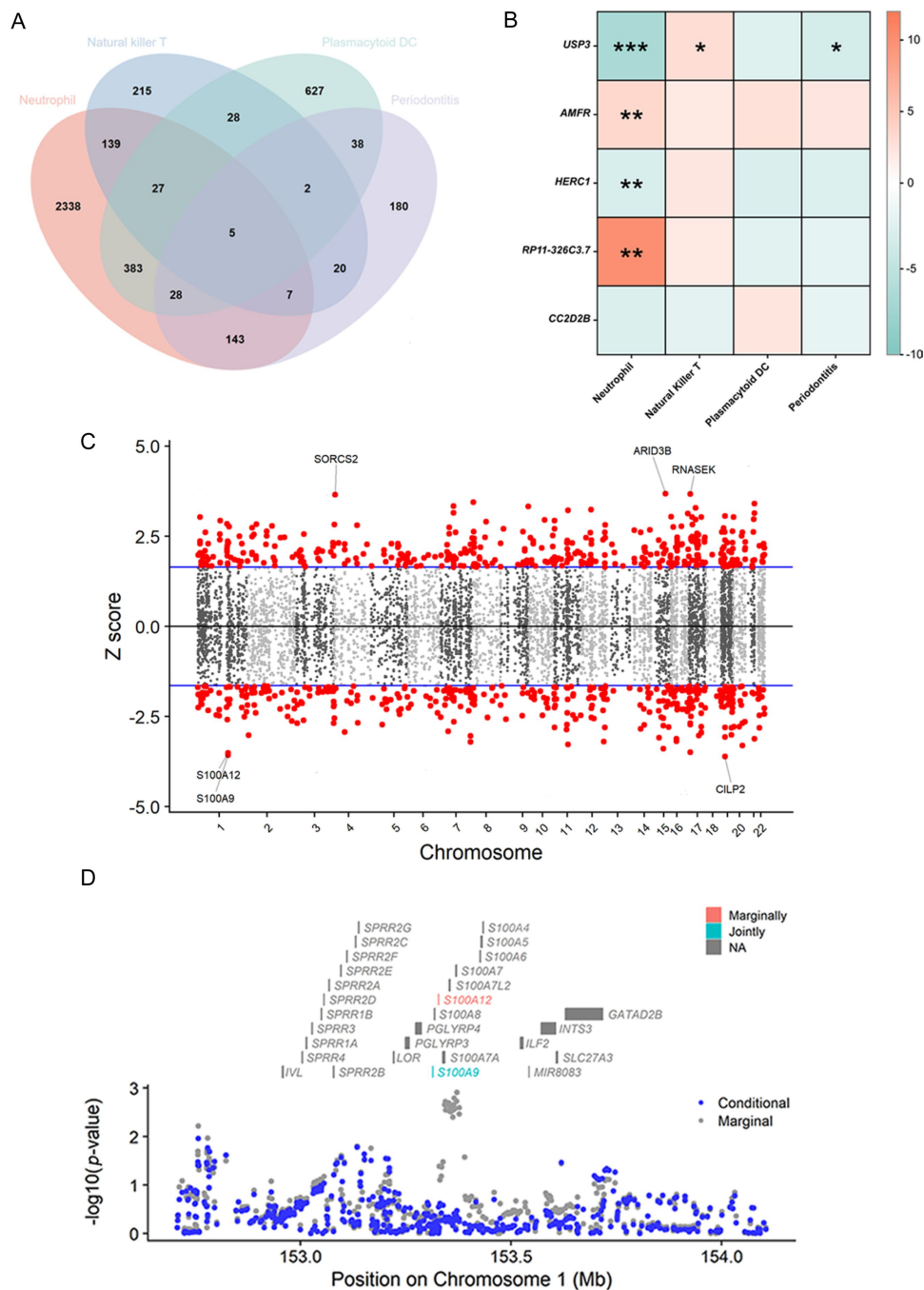


Figure 5.

Results of the TWAS and colocalization analysis. **(A)** A Venn diagram illustrates the intersecting genes shared by multiple traits ($P < 0.05$). **(B)** A heatmap representing the TWAS and colocalization analysis for five genes interacting among neutrophil, Natural Killer T cell, plasmacytoid DC, and periodontitis. The TWAS Z-score is used as the color scale for the heatmap. * PP.H3 + PP.H4 > 0.5; ** PP.H3 + PP.H4 > 0.8; *** PP.H4 > 0.8. **(C)** Manhattan plot of gene-trait associations for periodontitis. The X-axis represents genomic positions. Blue lines indicate a Z-score of 1.96. Red circles represent significant gene-trait associations ($P < 0.05$). Six genes satisfy a multiple corrected threshold of $P < 5 \times 10^{-4}$. **(D)** Regional Manhattan plot of conditional analysis for S100A9, S100A12 in periodontitis. Grey bars indicate the location of genes on chromosome 1. Genes colored in orange and green on the graph indicate the marginally and jointly significant genes that best explain the GWAS signals. Gray and blue dots respectively indicate GWAS p-values before and after conditioning on the jointly significant gene. **Abbreviations:** DC, dendritic cell; GWAS, genome-wide association study; PP, posterior probability; TWAS, transcriptome-wide association study.

Gene name	Position	Phenotype	TWAS Z-score	TWAS <i>p</i> val	Perm <i>p</i> val	Model <i>p</i> val	PP.H3	PP.H4
<i>S100A9</i>	1q21.3	Neutrophil	7.225	5.0×10^{-13}	NA	1.7×10^{-17}	0.035	0.962
		Periodontitis	-3.564	3.7×10^{-4}	1.3×10^{-3}	1.7×10^{-17}	0.008	0.259
<i>S100A12</i>	1q21.3	Neutrophil	7.310	2.7×10^{-13}	1.7×10^{-4}	4.1×10^{-43}	0.027	0.970
		Periodontitis	-3.513	4.4×10^{-4}	8.1×10^{-4}	4.1×10^{-43}	0.008	0.244
<i>MCM6</i>	2q21.3	Neutrophil	-5.549	2.9×10^{-8}	NA	3.5×10^{-13}	0	0.999
		Plasmacytoid DC	7.172	7.4×10^{-13}	2.7×10^{-3}	3.5×10^{-13}	1.000	0
<i>PI4KAP2</i>	22q11.21	Neutrophil	5.345	9.0×10^{-8}	3.3×10^{-3}	3.7×10^{-39}	0.003	0.016
		Plasmacytoid DC	5.322	1.0×10^{-7}	3.2×10^{-3}	3.7×10^{-39}	0.009	0.045

Abbreviations: DC, dendritic cell; Perm, permutation test; PP, poster probability; TWAS, transcriptome-wide association study.

Table 3.

TWAS and colocalization analysis identified genes involved in multiple phenotypes.

([Brito et al., 2013](#)). The quantity and functionality of neutrophils both act as critical indicators of inflammation severity. The reduction in neutrophil count and inflammatory mediators, observed after successful periodontitis treatment, suggests a reduction in systemic inflammation ([Hajishengallis, 2022](#)).

The recently developed concept of "trained immunity" has introduced new perspectives in understanding how neutrophils contribute to periodontitis and its comorbidities. "Trained Immunity" primarily denotes the memory-like response of myeloid innate immune cells, such as neutrophils and monocytes, to pathogens following an initial infection ([Li et al., 2022](#)). Contrary to the memory response of specific immune cells, the memory characteristic of trained immunity primarily manifests as a rapid response and efficient elimination of pathogens, rather than specific recognition ([Netea et al., 2016](#)). Trained myeloid cells have the potential to amplify the functionality of neutrophils, thereby fortifying the body's defense against subsequent infections. Nevertheless, within the framework of chronic inflammation, these cells could potentially intensify tissue damage ([Hajishengallis and Chavakis, 2022](#)).

Periodontitis exemplifies a condition that stimulates maladaptive myelopoiesis in the bone marrow, characterized by the production of an excessive number of hyper-inflammatory monocytes and neutrophils ([Li et al., 2023](#)). This leads to an influx of maladaptively trained neutrophils in both the blood circulation and periodontal tissues, that infiltrate not only oral tissues but also non-oral ones, simultaneously triggering an increase in the production of Neutrophil Extracellular Traps (NETs) and a reduction in their degradation ([White et al., 2016](#)). This process, as a result, intensifies the collapse of the epithelial barrier, fosters bacteremia, exacerbates periodontitis, and amplifies the severity of inflammatory complications ([Burmeister et al., 2022](#)). While our comprehension of the regulatory mechanisms and functions of neutrophils is far from complete, current research has facilitated the development of targeted therapeutic strategies to manage chronic inflammatory disorders mediated by neutrophils, such as the inflammatory response observed in periodontitis.

Several lymphocyte subsets causally associated with the risk of periodontitis

NKT cells, a distinct fraction of T lymphocytes, are linked to the pathophysiology of a variety of inflammatory, osteolytic, and autoimmune diseases ([Godfrey et al., 2000](#)). Similar to our findings, previous research revealed a greater number of NKT recruited in periodontitis tissues ([Muthukuru, 2012](#); [Yamazaki et al., 2001](#)). Several studies have demonstrated the tissue-specific function of NKT and highlighted its pathogenic role in periodontitis ([Aoki-Nonaka et al., 2014](#); [Melgar-Rodríguez et al., 2021](#)), which may be attributed to the proinflammatory and immunoregulatory activities mediated by NKT, spanning from cytokine production to immune cell interactions ([Seidel et al., 2020](#)).

In addition, our study identified a convoluted causal relationship between pDC and periodontitis. Dendritic Cells, as specialized antigen-presenting cells, play a crucial role in the modulation of the host immune response and may be related to bone loss during periodontitis ([El-Awady et al., 2022](#); [Ginesin et al., 2023](#)). In response to viral encounters and infection, pDC represents a unique subgroup of DC that releases type I interferon (IFN) ([Jego et al., 2003](#)). However, pDC is only discovered in a tiny percentage of healthy oral tissues, and there remains a dearth of relevant clinical research ([Meghil and Cutler, 2020](#); [Wilensky et al., 2014](#)). The involvement of pDC in periodontitis deserves further investigation.

Systemic immunomodulation management for immune cells serves as a target for periodontal care

Periodontitis is a damaging inflammatory disease induced and exacerbated by the plaque biofilm and host immune response ([Moutsopoulos and Konkel, 2018](#)). The systemic immune response comprises both innate and adaptive immunity, with numerous cytokines, immune cells, and inflammatory pathways interacting in a complex crosstalk during periodontitis, hinting that immunomodulation management may be an essential target for periodontal care ([Dutzan et al., 2016](#); [Hajishengallis, 2014](#)).

Reactive periodontal therapies, which focus on plaque management, pocket depth reduction, and gingival bleeding eradication, do not always produce the intended results and fall short of a genuinely comprehensive approach to dental care ([Kornman et al., 2017](#)). Recently, a promising term "P4 periodontics" (Predictive, Preventive, Personalized, and Participatory) has been introduced as a multilayer healthcare paradigm for the management of periodontitis, emphasizing the personalized responsiveness of treatment to disease ([Bartold and Ivanovski, 2022](#)). Modulation of systemic host immune responses is particularly appropriate for predicting the progression and severity of periodontitis in persons whose periodontal condition is only slightly correlated with dental plaque ([Divaris et al., 2020](#)). MR contributes a novel approach to the investigation of systemic immunological alternations in periodontitis. A recent MR study evaluated the causal associations between circulating cytokines and the risk of periodontitis ([Huang et al., 2023](#)).

Our present study highlighted five genes (*USP3*, *AMFR*, *HERC1*, *CC2D2B*, and *RP11-326C3.7*) that may play a pivotal role in the communication between circulating neutrophils, pDC, NKT, and periodontitis, as well as two high confidence genes (*S100A9*, *S100A12*) situated within 1q21.3 as prospective gene targets for regulating circulating neutrophils during periodontitis. Our findings could pave the way for a novel preventive and therapeutic approach to modifying the systemic immunological equilibrium in periodontitis patients by modulating circulating immune cells. These findings may enable the prediction of individuals at risk of periodontitis through screening certain immune imbalances, which might then be employed to specifically prevent periodontitis and related inflammatory comorbidities, particularly in patients with systemic susceptibility factors.

Strengths and limitations in the present study

The present study exhibited several strengths. First, under the premise of three key assumptions, MR is a powerful tool for explaining the relationship between complicated features (such as circulating immune cells) by successfully mitigating the effect of probable confounders and allowing for reasonable causal order. Second, a rigorous quality control process was conducted in accordance with the STROBE-MR checklist in multiple domains, including IVs selection, heterogeneity investigations, and removal of pleiotropic loci. Third, we adopted a series of sensitivity tests and MVMR to rule out the impact of outlier, influential, or pleiotropic SNPs. Fourth, a novel method based on nonlinear Bayesian averaging was applied to explore the causal drivers of disease risk from a set of high-throughput risk factors. Finally, TWAS was used in conjunction with MR to identify achievable regulatory gene targets for periodontal care.

However, some limitations should be addressed when interpreting the results. To begin with, a scarcity of GWAS databases hampered more comprehensive and precise analyses. As a result, we were unable to evaluate the impact of immune cells on distinct subsets of periodontal illnesses (such as gingival recession, and periodontal abscess) or ethnic groups (such as East Asian and African). Second, the primary results from the IVW method were not stable across all alternative analyses, nor were they replicated within subgroups, implying that the findings had limited evidentiary power. Third, the considerable variation in sample size between the two exposure

databases contributes to the discrepancies in the number of positive SNPs. Despite our exploration of multiple selection thresholds for IVs, the inconsistency in screening methods and the discrepancy in the included SNPs could potentially introduce bias. Fourth, none of the results satisfied the Bonferroni correction ($P < 0.05/17$ traits = 0.003), which may have inflated the rate of type I errors. Fifth, in spite of our best attempts to minimize potential confounding factors, interferences from unobserved pleiotropies could not be completely ruled out. Fifth, we relied on the whole blood data For FUSION algorithm due to the lack of transcriptome data associated with oral tissues (such as gums, oral mucosa, and alveolar bone) in the GTEx database. This has led to an excessive focus on systemic immunological changes, thereby overlooking the significance of alterations in local periodontal tissue immunity. Such an oversight could potentially compromise the precision and pertinence of our research findings. Sixth, in addition to quantities, function abnormalities (such as dysregulation or hyperactivity) of circulating immune cells may also be related to the susceptibility and severity of periodontitis, however our research failed to address this issue. Seventh, while the majority of immune cells in gingival crevicular fluid are derived from blood, the amount of circulating immune cells is influenced by more intricate factors, which may challenge the current causal inference. Finally, since MR evaluates causal inference from the standpoint of genetic variations, it may not always correspond exactly to fact.

Conclusions

In conclusion, the present study provides suggestive evidence of the causal associations of genetically predicted circulating neutrophils, Natural Killer T cells, and plasmacytoid Dendritic Cells on the risk of periodontitis, which shed light on the involvement of systemic immunological alterations in periodontitis etiology. Our findings may provide an innovative and evidence-based framework for the prospect of systemic immunomodulation management in periodontal care, which can be valuable for early diagnostics, risk assessment, targeted prevention, and personalized management of periodontitis, especially for patients with systemic susceptibility factors. However, the effect estimation discovered in our study was marginal, prompting caution when transferring to clinical practice. More studies are required to comprehend the biological plausibility and clinical applicability of our findings.

Data Availability

The data generated or analysed during this study are available in this published article and its supplementary information files. The data used in the study could be requested through the corresponding author with reasonable request.

<https://data.bris.ac.uk/data/dataset/>

<https://finngen.gitbook.io/documentation/v/r9/data-download>

<https://gtexportal.org/home>

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Additional information

Competing interests

The authors declare no competing interests.

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Ethics

Ethics approval and consent to participate are not applicable

Additional files

Supplementary files

Supplementary File 1. Tables s1-16.

Table s1. Case definition and exclusion criteria in included GWASs.

Table s2. Characteristics of the SNPs used as IVs in the MR.

Table s3. Effect estimates of causal associations between circulating immune cells and periodontitis risk.

Table s4. The heterogeneity and horizontal pleiotropy of the main results using Cochran's Q statistics and MR-Egger intercept.

Table s5. Effect estimates of causal associations after excluding outliers in RadialMR for two features with considerable heterogeneity.

Table s6. Effect estimates of causal associations after excluding influential or outlier SNPs for significant results.

Table s7. Effect estimates of causal associations after mutual and covariate correction in MVMR.

Table s8. Ranking of models according to their PP using MR-BMA analysis.

Table s9-11. Effect estimates of causal associations between circulating immune cells and chronic periodontitis, chronic gingivitis, and gingival hyperplasia in subgroup analysis.

Table s12. Effect estimates of the reverse associations between periodontitis and circulating immune cells.

Table s13-16. Results of TWAS, conditional analysis, permutation testing and colocalization analysis on neutrophil, Natural Killer T cell, plasmacytoid DC and periodontitis.

Supplementary File 2. Figures s1-5.

Figure s1. Scatter plot to explore outliers for two features with considerable heterogeneity.

Figure s2. Results of leave-one-out sensitivity analysis.

Figure s3. Scatter plot to explore outliers for significant results.

Figure s4. Scatter plot of the Cochran's Q test and Cook's distance to explore outlier or influential variations in MR-BMA.

Figure s5. Results of the TWAS and colocalization analysis.

Supplementary File 3. STROBE-MR Checklist.

Data availability

Data from GLIDE can be obtained via application (<https://data.bris.ac.uk/data/dataset/>), FinnGen can be obtained via application (<https://finngen.gitbook.io/documentation/v/r9/data-download>), and GTEx can be obtained via application (<https://gtexportal.org/home>). The data generated or analysed during this study are available in this published article and its supplementary information files. The data used in the study could be requested through the corresponding author with reasonable request.

Code availability

The software and code used in the study could be obtained through the corresponding author.

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Reviewer #1 (Public Review):

Ye et al. used Mendelian randomization method to evaluate the causative association between circulating immune cells and periodontitis, and finally screened out three risk immune cells related to periodontitis. Overall, this is an important and novel piece of work that has the potential to contribute to our understanding of the causal relationship between circulating immune cells related to periodontitis.

<https://doi.org/10.7554/eLife.92895.2.sa1>

Reviewer #2 (Public Review):

Summary:

This is a carefully done study containing interesting results.

Strengths:

These findings have significant implications for periodontal care and highlight the potential for systemic immunomodulation management on periodontitis, which is of interest to readers in the fields of periodontology, immunology, and epidemiology.

<https://doi.org/10.7554/eLife.92895.2.sa0>

Author Response

The following is the authors' response to the original reviews.

eLife assessment

The authors analyzed the causative association between circulating immune cells and periodontitis, and reported three risk immune cells related to periodontitis. The significance of the findings is fundamental, which substantially advances our understanding of periodontitis. The strength of evidence is convincing.

Reviewer #1 (Public Review):

Ye et al. used Mendelian randomization method to evaluate the causative association between circulating immune cells and periodontitis and finally screened out three risk immune cells related to periodontitis. Overall, this is an important and novel piece of work that has the potential to contribute to our understanding of the causal relationship between circulating immune cells related to periodontitis. However, there are still some concerns that need to be addressed.

We sincerely appreciate the constructive feedback from the editor and reviewers, which has been instrumental in enhancing the quality of our manuscript.

(1) The authors used 1e-9 as the threshold to select effective instrumental variables (IVs), which should give the corresponding references. Meanwhile, the authors should test and discuss the potential impact of inconsistent thresholds for exposure (1e-9, 5e-6 were selected by the author respectively) and outcome IVs (5e-8) on the robustness of the results.

Thank you for your insightful comments. We have selected two GWAS databases as the data sources for the exposure group: the BCC Consortium with a sample size of 563,946, and the Sardinian cohort of 3,757. The considerable disparity in sample size between them may result in variations in outcomes, primarily showcased in the differences in positive SNP numbers. We, therefore, adopted an unconventional (non 5e-8) yet rigorously controlled screening strategy, an approach that is widely accepted in MR studies (Li et al., 2022; Liu et al., 2023). We believe that the present thresholds are sufficiently rigorous to guarantee the validity of the subsequent Mendelian randomization analysis.

However, employing two distinct methods in exposure screening is not typical, and we posit that this method can be viewed as an innovative strategy, providing a reference for future research dealing with two databases with significant discrepancies (Huang et al., 2023; Kong et al., 2023). As you perceptively noted, we acknowledge that this strategy may exert a certain influence on the research outcomes, and we have factored this potential limitation into our manuscript. "Third, the considerable variation in sample size between the two exposure databases contributes to the discrepancies in the number of positive SNPs. Despite our exploration of multiple selection thresholds for IVs, the inconsistency in screening methods and the discrepancy in the included SNPs could potentially introduce bias." (Page 14)

As for the "outcome IVs with 5e-8" you mentioned, we didn't implement this screening threshold in the outcome IVs. Indeed, we applied the same screening criteria as specified at 5e-06 (refer to Stable 2). Is the statement that you're referring to the following: "Additionally, SNPs that displayed a direct association with the outcome would also be excluded to uphold the third MR assumption ($P < 5e-8$)" (Page 6)? In this context, we adopted a standard criterion in the IVs screening process to remove SNPs directly associated with the outcome.

Reference

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(2) *What is the reference for selecting Smoking, Fasting plasma glucose, and BMI as covariates? They do not seem to be directly related to immune cells as confounding factors.*

The variables of Smoking, Fasting Plasma Glucose (FPG), and Body Mass Index (BMI) are commonly used as covariates in multivariable Mendelian randomization studies (Kong et al., 2023; Liu et al., 2023). The association between Smoking, FPG, and BMI with immune cells may not be immediately apparent. However, these factors have been identified as potential confounders that could impact overall health, which in turn may indirectly modulate systemic immune responses, susceptibility, and inflammation.

(1) . Smoking: It has been well-documented that smoking can cause inflammation and impair immune function, thereby increasing individual's susceptibility to infections and diseases (Shiels et al., 2014). As such, smoking is recognized as a covariate that could potentially influence the outcomes of an investigation into immune cells.

(2) FPG: Elevated FPG levels indicate poor glycemic control, potentially leading to conditions like diabetes (Choi et al., 2018). Consequently, studies have demonstrated that elevated FPG levels can compromise the immune system's ability to combat infections.

(3) BMI: It is a measure of body fat that takes into account a person's weight and height. Both obesity, characterized by a high BMI, and underweight, characterized by a low BMI, have been associated with a range of health issues, inclusive of a compromised immune system (Piñeiro-Salvador et al., 2022). Consequently, BMI is factored in as a covariate in this study.

We have thus incorporated these factors as covariates in our study to mitigate their potential confounding effects. The selection of these covariates is primarily guided by previous research and established knowledge concerning the potential influences on immune function. We appreciate your query and will ensure to clarify this point in our revised manuscript. "We have incorporated covariates, including the number of cigarettes smoked, fasting plasma glucose (FPG) levels, and body mass index (BMI) into the MVMR analysis, given that these factors could indirectly affect systemic immune responses and inflammation (Liu et al., 2023)." (Page 6-7)

Reference

Choi S-C, Titov AA, Abboud G, Seay HR, Brusko TM, Roopenian DC, Salek-Ardakani S, Morel L. 2018. Inhibition of glucose metabolism selectively targets autoreactive follicular helper T cells. *Nat Commun* 9:4369. doi:10.1038/s41467-018-06686-0

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Chaturvedi AK. 2014. Cigarette smoking and variations in systemic immune and inflammation markers. J Natl Cancer Inst 106:dju294. doi:10.1093/jnci/dju294

(3) *It is not entirely clear about the correction of P-value for the total number of independent statistical tests.*

In our study, we used the Bonferroni correction to adjust the P-values for multiple comparisons. The adjusted P-value is calculated as the original P-value times the total number of independent statistical tests. Specifically, we applied multiple corrections in the following two aspects: First, we corrected the results of the FUSION algorithm in TWAS, with a correction value of $P < 6.27 \times 10^{-6}$ ($0.05/7,890$ genes) (Page 8). Second, we performed multiple corrections on the initial results of MR ($P < 0.05/17$ traits = 0.003). However, none of the results met the criteria after the correction, which is one of the limitations detailed in the discussion section of our study (Page 14).

(4) *The author used whole blood data to apply FUSION algorithm. Although whole blood is a representative site, the authors should add FUSION testing of periodontally relevant tissues, such as oral mucosa.*

We appreciate your insightful comments and suggestions. We concur that employing periodontally relevant tissues, like oral mucosa, for FUSION testing might yield more precise and pertinent results. However, in the Genotype-Tissue Expression project (GTEx) database, we could not find transcriptome data related to oral tissues, such as gums, oral mucosa, and alveolar bone (Review Table 1). Owing to the limitations of the database, in the context of our study, we primarily relied on whole blood data, given its availability and the extensive precedent documented in the literature for its utilization (Xu et al., 2023; Yuan et al., 2022).

We acknowledge that this is a limitation of our study and will certainly consider incorporating periodontally relevant tissues in our future research. In the revised manuscript, we have explicitly stated this limitation and underscored the necessity for additional studies to corroborate our findings with periodontally relevant tissues. Fifth, we relied on the whole blood data For FUSION algorithm due to the lack of transcriptome data associated with oral tissues (such as gums, oral mucosa, and alveolar bone) in the GTEx database. “Fifth, we relied on the whole blood data For FUSION algorithm due to the lack of transcriptome data associated with oral tissues (such as gums, oral mucosa, and alveolar bone) in the GTEx database. This has led to an excessive focus on systemic immunological changes, thereby overlooking the significance of alterations in local periodontal tissue immunity. Such an oversight could potentially compromise the precision and pertinence of our research findings.” (Page 15)

Author response table 1.

Organizations and Samplesize in the GTEx database

ID	Tissue	N	ID	Tissue	N
1	Muscle_Skeletal	706	28	Artery_Coronary	213
2	Whole_Blood	670	29	Brain_Cerebellum	209
3	Skin_Sun_Exposed_Lower_leg	605	30	Liver	208
4	Artery_Tibial	584	31	Brain_Cortex	205
5	Adipose_Subcutaneous	581	32	Brain_Nucleus_accumbens_basal_ganglia	202
6	Thyroid	574	33	Brain_Caudate_basal_ganglia	194
7	Nerve_Tibial	532	34	Brain_Cerebellar_Hemisphere	175
8	Skin_Not_Sun_Exposed_Suprapubic	517	35	Brain_Frontal_Cortex_BA9	175
9	Lung	515	36	Small_Intestine_Terminal_Ileum	174
10	Esophagus_Mucosa	497	37	Brain_Hypothalamus	170
11	Cells_Cultured_fibroblasts	483	38	Brain_Putamen_basal_ganglia	170
12	Adipose_Visceral_Omentum	469	39	Ovary	167
13	Esophagus_Muscularis	465	40	Brain_Hippocampus	165
14	Breast_Mammary_Tissue	396	41	Brain_Anterior_cingulate_cortex_BA24	147
15	Artery_Aorta	387	42	Cells_EBV-transformed_lymphocytes	147
16	Heart_Left_Ventricle	386	43	Minor_Salivary_Gland	144
17	Heart_Atrial_Appendage	372	44	Vagina	141
18	Colon_Transverse	368	45	Brain_Amygdala	129
19	Esophagus_Gastroesophageal_Junction	330	46	Uterus	129
20	Stomach	324	47	Brain_Spinal_cord_cervical_c-1	126
21	Testis	322	48	Brain_Substantia_nigra	114
22	Colon_Sigmoid	318	49	Kidney_Cortex	73
23	Pancreas	305	50	Bladder	21
24	Pituitary	237	51	Cervix_Endocervix	10
25	Adrenal_Gland	233	52	Cervix_Ectocervix	9
26	Spleen	227	53	Fallopian_Tube	8
27	Prostate	221	54	Kidney_Medulla	4

Reference

Xu J, Si H, Zeng Y, Wu Y, Zhang S, Shen B. 2023. Transcriptome-wide association study reveals candidate causal genes for lumbar spinal stenosis. *Bone Joint Res* 12:387–396. doi:10.1302/2046-3758.126.BJR-2022-0160.R1

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(5) The authors chose gingival hyperplasia as a secondary validation phenotype of periodontitis in this study. However, gingival recession, as another important phenotype associated with periodontitis, should also be tested and discussed.

We appreciate your insightful feedback highlighting the significance of incorporating gingival recession as a phenotype in periodontitis studies. Our emphasis on gingival hyperplasia in the study was primarily dictated by the initial study design and the data available from FinnGen R9K11. Notwithstanding the lack of gingival recession data in the available databases, we identified chronic gingivitis data in an earlier version of the Finnish database (FinnGen R5K11) as an alternative. We performed a Mendelian Randomization analysis on this dataset, with the results integrated into Supplementary Table 10. Concurrently, Table 1, Supplementary Table 1, Figure 4, and the corresponding descriptions in the manuscript were updated. We trust this adjustment can address the limitations identified in our research. We are confident that this not only augments the comprehensiveness of our study but also fosters a more holistic comprehension of periodontal disease.

(6) This study used GLIDE data as a replicated validation, but the results were inconsistent with FinnGen's dataset.

Thank you for your insightful comments and for bringing this issue to our attention. Indeed, it is of utmost importance to ensure the validity and reliability of our findings across various datasets. The observed inconsistency between the GLIDE data and FinnGen's dataset could be attributed to several reasons.

Firstly, this discrepancy might originate from the differences in population composition. The former is grounded on a comprehensive meta-analysis of cohorts focusing on periodontitis, whereas the latter utilizes a dataset from a full-phenotype cohort. In the former, the ratio of periodontitis to the control groups is approximately 1:2. In contrast, the ratio in the latter seems to be minuscule. The sample size in the FinnGen data may not suffice to detect the effects observed in the GLIDE dataset, given that larger exposure sizes enhance the ability to detect genuine associations.

Moreover, the heterogeneity of periodontitis can potentially result in variable outcomes. Phenotypic definition methods differ between the two databases. The GLIDE database diagnoses based on the criteria of Centers for Disease Control and Prevention/American Academy of Periodontology (CDC/AAP) and Community Periodontal Index (CPI) for physical signs. While the FinnGen database adopts the International Classification of Diseases (ICD) 10 standard for a comprehensive diagnosis. The former database employs a more practical yet broader standard for periodontitis, which might encompass pseudo-periodontitis.

Finally, the observed differences could be attributed to the variations in immune responses at distinct stages of periodontitis. During the initial stages of periodontitis, neutrophils and macrophages primarily mediate the immune response. With the progression of the disease, the involvement of T cells and B cells increases, thereby leading to a more intricate immune response (Darveau, 2010). Besides, the immune system's response to these oral health conditions is not uniform and can be influenced by multiple factors, including the individual's overall health, genetics, and lifestyle, potentially impacting the results (Hung et al., 2023).

Reference

Darveau RP. 2010. Periodontitis: a polymicrobial disruption of host homeostasis. *Nat Rev Microbiol* 8:481–490. doi:10.1038/nrmicro2337

Hung M, Kelly R, Mohajeri A, Reese L, Badawi S, Frost C, Sevathas T, Lipsky MS. 2023. Factors Associated with Periodontitis in Younger Individuals: A Scoping Review. *J Clin Med* 12:6442. doi:10.3390/jcm12206442

Reviewer #2 (Public Review):

This manuscript presents a well-designed study that combines multiple Mendelian randomization analyses to investigate the causal relationship between circulating immune cells and periodontitis. The main conclusions of the manuscript are appropriately supported by the statistics, and the methodologies used are comprehensive and rigorous.

These findings have significant implications for periodontal care and highlight the potential for systemic immunomodulation management on periodontitis, which is of interest to readers in the fields of periodontology, immunology, and epidemiology.

We greatly appreciate the positive feedback and valuable insights provided by the reviewer, which have significantly contributed to the improvement of our manuscript.

Reviewer #2 (Recommendations for The Authors):

**Abstract*

Line 30-32: "Two-sample bidirectional univariable MR followed by sensitivity testing, multivariable MR, subgroup analysis, and the Bayesian model averaging (MR-BMA) were performed to explore the causal association between them." What does the term "them" refer to here, please clarify it. The research method here is unclear, please reorganize it.

Line 39: "S100A9 and S100A12" here should be italic.

We appreciate your meticulous suggestions and have revised the methods section accordingly. Additionally, the two genes have been highlighted in italics for emphasis.

"Univariable MR, multivariable MR, subgroup analysis, reverse MR, and Bayesian model averaging (MR-BMA) were utilized to investigate the causal relationships. Furthermore, transcriptome-wide association study (TWAS) and colocalization analysis were deployed to pinpoint the underlying genes." (Page 1)

Introduction

Line 78-80: "As reported, the number of immune cells in periodontal tissue changes as periodontitis progresses, featuring an increase in monocytes, and B cells and a decrease in T cells." Does the author mean that both monocytes and B cells increase as periodontitis progresses?

We are grateful for your meticulous reading and perceptive inquiries. We would like to confirm the accuracy of your understanding. In lines 78-80, our intended message was to communicate that with the progression of periodontitis, there is an increase in both monocytes and B cells in the periodontal tissue. This represents a typical immune response to the infection, where these cells play a pivotal role in counteracting periodontal pathogens. To enhance clarity, we have revised these lines in the manuscript as follows:

"With the progression of periodontitis, there is a significant alteration in the quantity of immune cells present within the periodontal tissue. Specifically, an increase in the count of both monocytes and B cells is observed, whereas a decrease is noted in the count of T cells." (Page 3)

Method

Line 164-165: "As the main test, the MVMR-IVW method, offered by the MVMR-least absolute shrinkage and selection operator (MVMR-LASSO), and the MVMR-Egger method were chosen." The author's expression here is ambiguous.

In response to your comment on the ambiguity in lines 164-165, we have revised the sentence for clarity. We hope this addresses your concern and clarifies our point more effectively.

"The MVMR-IVW method was utilized as the primary test, supplemented by the MVMR-least absolute shrinkage and selection operator (MVMR-LASSO) and the MVMR-Egger method." (Page 7)

Table 1: FinnGen has a greater sample size and more SNPs than GLIDE; why do authors choose the latter as the primary analysis?

Our choice to utilize GLIDE as the primary analysis tool, instead of FinnGen, was mainly guided by the specific research question we aimed to address. Despite FinnGen offering a larger sample size and more SNPs, GLIDE offers a more specialized and targeted dataset that suits the unique requirements of our study. In most MR studies, a similar strategy is adopted, wherein a large database of disease GWAS meta is utilized for exploration, followed by validation in full phenotype cohort (such as UKBiobank and FinnGen) (Liu et al., 2023; Yuan et al., 2023). To summarize, the reasons may primarily include the following:

Firstly, GLIDE offers a concentrated and targeted methodology for examining genetic data pertinent to periodontitis. This dataset is grounded in a comprehensive meta-analysis of cohorts centered on periodontitis, wherein the ratio of periodontitis cases to control groups is approximately 1:2. Conversely, the proportion in FinnGen seems to be negligible, given that it employs a dataset derived from a comprehensive phenotype cohort. Consequently, employing the GLIDE database as a primary investigative tool can generate more abundant genetic information associated with periodontitis.

Furthermore, the methodological facets of GLIDE align more accurately with the analytical framework of our study. For instance, the diagnostic criteria methods vary between the two databases. The GLIDE database derives its basis from the Centers for Disease Control and Prevention/American Academy of Periodontology (CDC/AAP) and Community Periodontal Index (CPI) for physical indicators. In contrast, the FinnGen database employs the International Classification of Diseases (ICD) 10 standard for an exhaustive diagnosis. The former adopts a more pragmatic, yet broader, standard for diagnosing periodontitis. The latter continues to use concepts of diseases such as "chronic periodontitis", which have been replaced by "periodontitis" in the latest disease classification from the "2017 World Workshop on the Classification of Periodontal and Peri-Implant Diseases and Conditions" in the periodontal field (Caton et al., 2018).

Reference

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Result

Line 224: "The observed significant results remained robust after removing pleiotropic SNPs." It is not clear what the authors mean by "remain robust".

Line 229-231: "The causal relationship between neutrophils and periodontitis remained stable with no evidence of heterogeneity or pleiotropy." It is also not clear what the authors mean by "remain stable". How does the author get to the conclusion that there is no evidence of heterogeneity or pleiotropy?

Figure S5: Please offer a brief explanation on how to investigate outlier or influential changes using scatter plots and Cochran's Q test and Cook's distance.

Line 224: We apologize for the confusion caused by the term "remain robust". In the revised manuscript, we clarified this by stating, "The observed significant results are considered 'robust' if the effect of sensitivity analyses was identical to that of Inverse Variance Weighted (IVW) method, yielding a P-value less than 0.05." (Page 6)

Line 229-231: We used the terms "remain stable" and "remain robust" interchangeably to express the same idea. To clarify, we have now unified the expression in the revised manuscript. As for the conclusion of "no evidence of heterogeneity or pleiotropy", it is derived from the results of Cochran's Q and Egger's intercept tests ($P < 0.05$). We have added this explanation to the revised manuscript for better clarity.

Figure S5: In the revised manuscript and Table, we have provided a succinct explanation regarding the investigation of outliers or influential changes as follows: "A genetic variant was defined as either an outlier or an influential variant if it possessed a q-value exceeding 10 or if its Cook's distance surpassed the median of the corresponding F-distribution." (Page 7)

We have made all the necessary changes in the revised manuscript based on your comments. We hope our responses and revisions adequately address your concerns.

Discussion

I have consulted several pieces of literature to ensure a thorough explanation, which may be helpful for your writing.

(1) Hajishengallis G, Li X, Divaris K, Chavakis T. Maladaptive trained immunity and clonal hematopoiesis as potential mechanistic links between periodontitis and inflammatory comorbidities. Periodontol 2000. 2022;89(1):215-230. doi:10.1111/prd.12421

(2) Hajishengallis G, Chavakis T. Mechanisms and Therapeutic Modulation of Neutrophil-Mediated Inflammation. J Dent Res. 2022;101(13):1563-1571. doi:10.1177/00220345221107602

We appreciate your valuable feedback and the additional references you provided to enrich our manuscript. Upon receiving your comments, we have meticulously reviewed and incorporated the suggested literature into our revised manuscript. These references have furnished insightful information, which has been assimilated into the revised manuscript (Page 12) to enhance the explanation of the mechanisms of neutrophil-mediated

inflammation and the potential association between periodontitis and inflammatory comorbidities.

"The quantity and functionality of neutrophils both act as critical indicators of inflammation severity. The reduction in neutrophil count and inflammatory mediators, observed after successful periodontitis treatment, suggests a reduction in systemic inflammation (Hajishengallis , 2022)." (Page 12)

"Trained myeloid cells have the potential to amplify the functionality of neutrophils, thereby fortifying the body's defense against subsequent infections. Nevertheless, within the framework of chronic inflammation, these cells could potentially intensify tissue damage (Hajishengallis and Chavakis, 2022)." (Page 12)